

(Figure 47.23). The work of Spemann and others demonstrated that the first two blastomeres of the frog embryo are **totipotent**, meaning that they can each develop into all the different cell types of that species.

In mammals, embryonic cells remain totipotent through the eight-cell stage, much longer than in many other animals. Recent work, however, indicates that the very early cells (even the first two) are not actually equivalent in a normal embryo. Rather, their totipotency when isolated

likely means that the cells can regulate their fate in response to their embryonic environment. Once the 16-cell stage is reached, mammalian cells are determined to form the trophoblast or the inner cell mass. Although the cells have a limited developmental potential from this point onward, their nuclei remain totipotent, as demonstrated in transplantation and cloning experiments (see Figures 20.17 and 20.18).

The totipotency of cells early in human embryogenesis is the reason why you or a classmate may have an identical

twin. Identical (monozygotic) twins result when cells or groups of cells from a single embryo become separated. If the separation occurs before the trophoblast and inner cell mass become differentiated, two embryos grow, each with its own chorion and amnion. This is the case for about a third of identical twins. For the rest, the two embryos that develop share a chorion and, in very rare cases where separation is particularly late, an amnion as well.

Regardless of how uniform or varied early embryonic cells are in a particular species, the progressive restriction of developmental potential is a general feature of development in all animals. In general, the tissue-specific fates of cells are fixed in a late gastrula, but not always so in an early gastrula. For example, if the dorsal ectoderm of an early amphibian gastrula is experimentally replaced with ectoderm from some other location in the same gastrula, the transplanted tissue forms a neural plate. But if the same experiment is performed on a late-stage gastrula, the transplanted ectoderm does not respond to its new environment and does not form a neural plate.

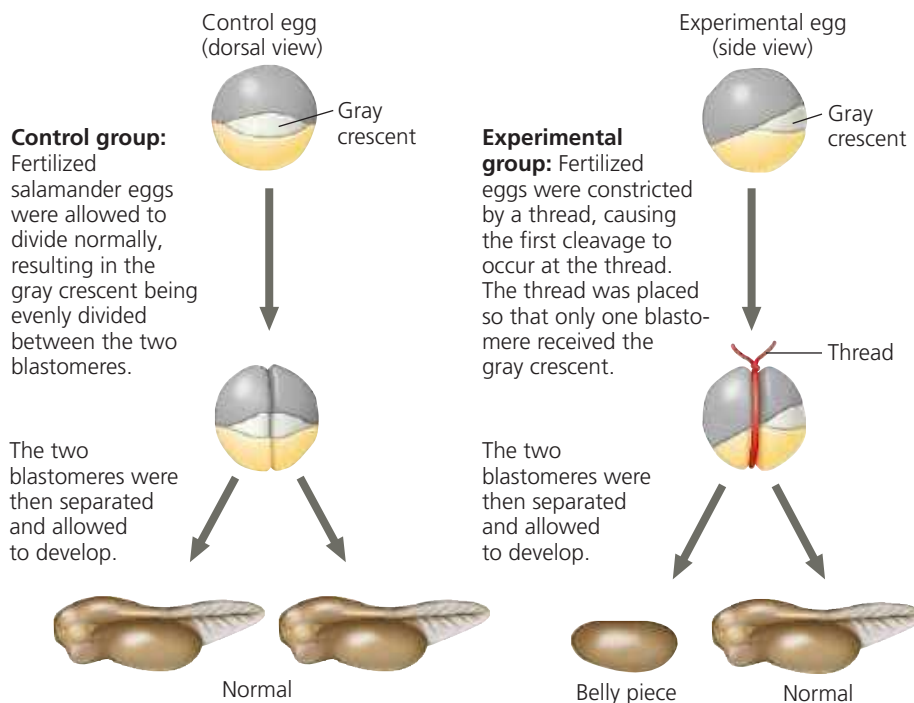
Cell Fate Determination and Pattern Formation by Inductive Signals

As embryonic development continues, cells influence each other's fates by induction. At the molecular level, the response to an inductive signal is usually to switch on a set of genes that make the receiving cells differentiate into a specific cell type or tissue. Here we will examine examples of this important developmental process in organizing the basic body plan of an embryo and in directing the three-dimensional development of a vertebrate limb.

▼ Figure 47.23 Inquiry

How does distribution of the gray crescent affect the developmental potential of the first two daughter cells?

Experiment Hans Spemann, at the University of Freiburg, Germany, carried out the following experiment in 1938 to test whether substances were located asymmetrically in the gray crescent.



Results Blastomeres that received half or all of the material in the gray crescent developed into normal embryos, but a blastomere that received none of the gray crescent gave rise to an abnormal embryo without dorsal structures. Spemann called it a "belly piece."

Conclusion The developmental potential of the two blastomeres normally formed during the first cleavage division depends on their acquisition of cytoplasmic determinants localized in the gray crescent.

Data from H. Spemann, *Embryonic Development and Induction*, Yale University Press, New Haven, CT (1938).

DRAW IT Draw lines to show the plane of the first cell division that would occur in the fertilized eggs above if no manipulations were done.

WHAT IF? In a similar experiment 40 years earlier, embryologist Wilhelm Roux allowed the first cleavage to occur and then used a needle to kill just one blastomere. The embryo that developed from the remaining blastomere (plus remnants of the dead cell) was abnormal, resembling a half-embryo. How might the presence of molecules in the dead cell explain why Roux's result differed from the control result in Spemann's experiment?